

An AI-Powered NGO Media Management & Curation WebApp

Submitted to

University of Mumbai

In partial fulfillment of the requirements for the degree of

Bachelor of Engineering

in

Computer Engineering

Khan Zaid Zakir (24CO55)

Macky Numan Nadeem (24CO56)

Masam Sachin Ramesh (24CO58)

Mirza Mohammed Ibrahim Baig (24CO59)

Shaan Ali Shaikh (24CO60)

Mohammed Anas Jamal Ahmed (24CO61)

Mohammed Arif Kazi (24CO62)

Mohammed Faizan Ateeq Ahmed (24CO63)

Patel Hamza M.Anjum (25DCO07)

Pawle Zainulabideen Muzaffar (25DCO08)

Supervisor

(Prof. Malik Sadaf)



Department of Computer Engineering

Anjuman-I-Islam's Kalsekar Technical Campus

School of Engineering and Technology

Plot No. 2 & 3, Sector -16, Near Thana Naka, Khanda Gaon,

New Panvel, Navi Mumbai. 410206

Academic Year: 2025-26

Anjuman-I-Islam's Kalsekar Technical Campus

School of Engineering and Technology

Department of Computer Engineering

Khanda Gaon, New Panvel, Navi Mumbai. 410206



Certificate

This is to certify that, the CEP Project entitled
“An AI-Powered NGO Media Management & Curation WebApp”

Khan Zaid Zakir (24CO55)

Macky Numan Nadeem (24CO56)

Masam Sachin Ramesh (24CO58)

Mirza Mohammed Ibrahim Baig (24CO59)

Shaan Ali Shaikh (24CO60)

Mohammed Anas Jamal Ahmed (24CO61)

Mohammed Arif Kazi (24CO62)

Mohammed Faizan Ateeq Ahmed (24CO63)

Patel Hamza M. Anjum (25DCO07)

Pawle Zainulabideen Muzaffar (25DCO08)

and is submitted in the partial fulfillment of the requirement for the degree of

Bachelor of Engineering

in

Computer Engineering

Submitted to

University of Mumbai



Project Guide
(Prof. Malik Sadaf)

Project Coordinator
(Prof. Nusrat Jahan)

Head of Department
(Prof. Tabrez Khan)

Dean (SoET)
(Dr. Rajendra Magar)

Community Engagement Project Approval

This Community Engagement Project entitled “ *An AI-Powered NGO Media Management & Curation WebApp* ” is a bonafide work done by *Khan Zaid Zakir (24CO55)*, *Macky Numan Nadeem (24CO56)*, *Masam Sachin Ramesh (24CO58)*, *Mirza Mohammed Ibrahim Baig (24CO59)*, *Shaan Ali Shaikh (24CO60)*, *Mohammed Anas Jamal Ahmed (24CO61)*, *Mohammed Arif Kazi (24CO62)*, *Mohammed Faizan Ateeq Ahmed (24CO63)*, *Patel Hamza M.Anjum (25DCO07)*, *Pawle Zainulabideen Muzaffar (25DCO08)* under the supervision of *Prof. Malik Sadaf*. This project is approved in the partial fulfillment of the requirement for the degree of *B.E. in Computer Engineering*.

Internal Examiner :

1.

2.

External Examiners :

1.

2.

Date : .../.../.....

Place :

Declaration

We declare that this written submission represents our ideas and does not involve plagiarism. We have adequately cited and referenced the original sources wherever others' ideas or words have been included. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action against us by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Khan Zaid Zakir (24CO55)

Macky Numan Nadeem (24CO56)

Masam Sachin Ramesh (24CO58)

Mirza Mohammed Ibrahim Baig (24CO59)

Shaan Ali Shaikh (24CO60)

Mohammed Anas Jamal Ahmed (24CO61)

Mohammed Arif Kazi (24CO62)

Mohammed Faizan Ateeq Ahmed (24CO63)

Patel Hamza M.Anjum (25DCO07)

Pawle Zainulabideen Muzaffar (25DCO08)

Date : .../.../.....

Abstract

This report documents the "Trustfeed" project, an AI-powered media management and curation Progressive Web App (PWA) developed for the Create Together Foundation. NGOs frequently struggle with fragmented media collection, high storage costs, and the administrative bottleneck of manually curating field photos for public relations. To address these issues, we developed a comprehensive solution utilizing a dual-storage architecture (Firebase for metadata and Cloudflare R2 for bulk media) and integrated the Google Gemini API for automated image tagging. The system consists of a mobile-first live feed for volunteers to securely upload media directly from the field, and a director dashboard that functions as a headless CMS, allowing one-click publishing of AI-drafted event summaries to a live public widget. This approach significantly reduces the manual workload of NGO directors, ensures high-quality media preservation without exorbitant costs, and provides donors with real-time, transparent updates on the organization's impact.

Contents

Abstract	i
List of Figures	ii
List of Tables	ii
1 Introduction	1
1.1 Overview	1
1.2 Motivation	1
1.3 Objectives	2
1.4 Organization of the Report	3
2 Community Study	4
2.1 Selected Community Details	4
2.2 Community Background and Operational Constraints	4
2.3 Detailed Observations and Workflow Analysis	5
2.3.1 Inefficiencies in Media Collection	6
2.3.2 The Financial Burden of Storage	6
2.3.3 The Administrative Curation Bottleneck	6
3 Problem Identification	8
3.1 Problem Statement	8
3.2 Detailed Analysis of Identified Issues	8
3.2.1 Fragmented Media Collection and Catastrophic Metadata Loss	8
3.2.2 Prohibitive and Unscalable Storage Costs	9
3.2.3 The Manual Curation and Publishing Bottleneck	9
3.3 The Critical Need for an Automated Solution	10
4 Methodology and Activities	11
4.1 Methodology and Implementation Strategy	11
4.2 System Architecture and Data Flow	12
4.3 Tools, Technologies, and Resources Utilized	13
4.4 User Interface Design and Implementation	13
5 Results and Impact	15
5.1 Technical Outcomes and System Performance	15
5.2 Quantifiable Impact on the Community Ecosystem	15

6	Conclusion and Future Work	18
6.1	Conclusion	18
6.2	Future Work and Scalability Roadmaps	18
	Acknowledgement	21

List of Figures

1.1	Conceptual diagram illustrating the data degradation and administrative bottle-neck inherent in relying on standard messaging applications for field data collection.	2
2.1	Images from the field visit and requirement gathering session with the Directors at the Create Together Foundation office.	5
2.2	Flowchart illustrating the data degradation and workflow friction in the existing ad-hoc messaging workflow utilized by the NGO.	6
4.1	Comprehensive System Architecture of the Trustfeed Platform, detailing the specific microservice interactions and the optimized data flow pipeline.	12
4.2	[Placeholder] The Volunteer Progressive Web App (PWA) Interface rendered on a standard mobile viewport, highlighting the native-app feel.	14
4.3	[Placeholder] The Director’s Curation Dashboard rendered on a high-resolution desktop viewport, detailing the AI tag review grid.	14
5.1	[Placeholder] The live Impact Widget dynamically embedded directly onto the NGO’s pre-existing public website, instantly surfacing AI-curated field data to donors.	17

List of Tables

- 2.1 Comprehensive comparison between the existing media collection method and the proposed Trustfeed architecture. 7
- 3.1 Estimated weekly administrative time savings realized by utilizing the highly automated Trustfeed platform. 10
- 4.1 Detailed summary of the core web technologies utilized in the Trustfeed full-stack architecture. 13

Chapter 1

Introduction

The management and curation of digital media have become significant, often overwhelming challenges for non-governmental organizations (NGOs) operating at the grassroots level. As these organizations scale their activities to address complex societal issues ranging from educational deficits to environmental degradation, the volume of data captured by volunteers in the field grows exponentially. This rapid influx of digital assets necessitates highly structured, scalable approaches to data storage, organization, and public presentation. Without robust digital infrastructure, the vital work performed by these organizations often goes unnoticed, hindering their ability to secure funding, recruit volunteers, and maintain transparency with their donors. This chapter establishes the fundamental premise of the project, detailing the core issues and outlining the proposed technological intervention.

1.1 Overview

This project, titled "Trustfeed," introduces a comprehensive, AI-powered media management and curation Progressive Web App (PWA). It directly addresses the real-time observation that NGOs struggle significantly with organizing and publishing field data. Daily, volunteers capture valuable photographs, videos, and documents that provide irrefutable evidence of their community impact. However, the efficient collection, secure storage, intelligent curation, and subsequent updating of public-facing platforms remain chaotic processes heavily reliant on manual human intervention. This system provides a streamlined, centralized platform acting as a sophisticated bridge between raw field data collection and polished, professional public relations outputs. By automating the most tedious aspects of media management—specifically ingestion and tagging—Trustfeed allows NGOs to focus their limited resources on their core philanthropic missions rather than administrative IT overhead. The resulting platform is inherently highly available, hardware-agnostic, and deeply integrated into the modern web ecosystem.

1.2 Motivation

The primary motivation for this project stems from a deep, systematic analysis of the high manual effort required by NGO directors and administrative staff to sort, curate, and upload media to organizational websites. This manual bottleneck consistently leads to delayed public relations updates, disorganized internal document repositories, and a general lack of visibility into the organization's daily operations.

Furthermore, the technical quality of media is often severely degraded due to the ubiquitous reliance on standard consumer messaging applications, such as WhatsApp or Telegram, for media transfers from the active field to the central office. These platforms employ aggressive, proprietary compression algorithms designed for quick transmission over cellular networks, which irreversibly degrades image resolution. Crucially, these platforms also systematically strip away valuable EXIF (Exchangeable Image File Format) metadata, including geospatial coordinates (GPS location), accurate capture timestamps, and device analytics. This complete loss of context makes it nearly impossible to accurately catalog historical data, verify the authenticity of field reports, or seamlessly organize media by event or location without intensive, error-prone manual data entry by the administrative staff. The motivation is to completely eradicate this loss of fidelity and context.

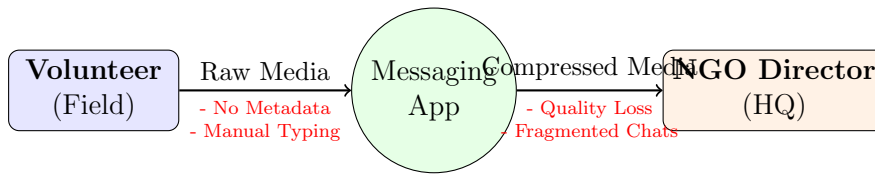


Figure 1.1: Conceptual diagram illustrating the data degradation and administrative bottleneck inherent in relying on standard messaging applications for field data collection.

1.3 Objectives

The overarching goal of this project is to architect and deploy a "High-Trust" Progressive Web App (PWA) to streamline batch media uploads directly from native smartphone OS galleries, completely circumventing quality degradation and metadata loss. The main objectives are delineated as follows:

1. **Secure Storage and Access Control:** Implement a robust authentication system utilizing JSON Web Tokens (JWT) and OAuth to securely separate private, sensitive administrative documents from public-facing feeds, ensuring data integrity, privacy, and strict compliance with modern data protection standards.
2. **Automated AI Tagging and Description Generation:** Seamlessly integrate the Google Gemini multimodal LLM API to automatically analyze image contents, generate relevant descriptive tags, and draft comprehensive PR summaries for uploaded media, thereby entirely removing the burden of manual metadata entry and copywriting from the administrative staff.
3. **Cost-Effective Dual-Storage Architecture:** Design a highly scalable cloud infrastructure that utilizes low-cost object storage (Cloudflare R2) for heavy, high-resolution media assets while maintaining a fast, globally distributed NoSQL database (Firestore) for relational metadata and application state.

4. **Seamless Publishing Workflow:** Develop a dedicated, framework-agnostic JavaScript widget that acts as a headless Content Management System (CMS), allowing directors to easily review AI-generated content and publish curated media directly to their existing legacy HTML website with a single click.

1.4 Organization of the Report

The remainder of this report is organized to provide a comprehensive, logical progression through the project’s engineering lifecycle. Chapter 2 details the community study, focusing on the selected NGO and their specific operational challenges, providing the necessary context for the system design. Chapter 3 dives deeply into defining the system’s core issues and practically justifying the need for the proposed solution. Chapter 4 provides an extensive, highly technical explanation of the methodology, system architecture, and the modern web technologies utilized for the full-stack implementation. Chapter 5 discusses the quantifiable outcomes and the profound impact the system has had on the community ecosystem. Finally, Chapter 6 concludes the comprehensive report, summarizing the achievements and proposing detailed avenues for future enhancements and scalability.

Chapter 2

Community Study

The primary objective of this chapter is to deeply introduce the community organization that served as the catalyst for the identification and subsequent selection of this engineering task. It comprehensively outlines the background of the selected Non-Governmental Organization (NGO), their daily operational constraints, and the crippling limitations of their pre-existing digital workflow. The general observations derived from this extensive community study unequivocally justify the absolute necessity for a tailored, AI-powered media management solution designed specifically for resource-constrained, high-impact environments.

2.1 Selected Community Details

The community organization selected as the primary stakeholder, testing environment, and ultimate beneficiary for this project is the **Create Together Foundation**. Co-directed by Deepak Vishwakarma and Siddhesh Lokhare, this NGO operates extensively primarily out of Kalamboli, Panvel, with a wide, continuously expanding reach across the surrounding suburban and rural districts of Maharashtra. Their organizational mandate focuses heavily on sustained community service, specifically focusing on transforming education in Zilla Parishad (ZP) schools and executing Women Empowerment initiatives (such as recycling waste clothes into reflective dog collars to prevent accidents). Recently, they successfully executed "Mission 3033," a massive public funding campaign that raised over 3 Crores in 45 days to digitalize 30 ZP schools across the state. To execute these far-reaching programs effectively, the foundation relies heavily on a decentralized network of highly dedicated, albeit technically diverse, volunteers to execute ground-level tasks and physically track the impact of their multifaceted initiatives.

2.2 Community Background and Operational Constraints

Organizations operating in the non-profit sector, much like the Create Together Foundation, perpetually operate within extremely constrained financial and human resource environments. While they rely almost entirely on their volunteer base to document their fieldwork, distribute resources, and visually track the tangible impact of their initiatives, they severely lack dedicated, full-time Information Technology (IT) staff.

Furthermore, their operational budgets are allocated primarily to direct community aid and logistical necessities, leaving negligible room to afford expensive, enterprise-grade media management software or digital asset management (DAM) systems like those utilized by corporate



Figure 2.1: Images from the field visit and requirement gathering session with the Directors at the Create Together Foundation office.

entities. Consequently, their technical infrastructure for handling digital assets has evolved in a highly rudimentary, reactive, and ad-hoc manner. Their existing systems—comprising fragmented local hard drives, overloaded email inboxes, and free-tier cloud accounts—are fundamentally incapable of scaling efficiently to meet the demands of their increasing volume of community activities. This results in massive data silos, lost historical records, and a chronic inability to effectively communicate their successes to potential philanthropic donors or government grant agencies.

2.3 Detailed Observations and Workflow Analysis

During our prolonged engagement and study of the Create Together Foundation, we conducted extensive interviews with key stakeholders and mapped their entire data lifecycle from field capture to website publication. Several critical observations were made regarding their deeply flawed media handling processes:

2.3.1 Inefficiencies in Media Collection

Volunteers operating in the field typically resort to using generic, consumer-grade messaging applications (predominantly WhatsApp and Telegram) to send event photos and videos back to the central administration. This process is inherently flawed for professional archiving. These platforms aggressively strip images of crucial EXIF metadata—most notably the precise GPS coordinates and the original capture timestamps. Furthermore, they apply heavy lossy compression algorithms to save bandwidth, resulting in poor visual quality that is often entirely unusable for high-resolution print campaigns, annual reports, or professional public relations materials.

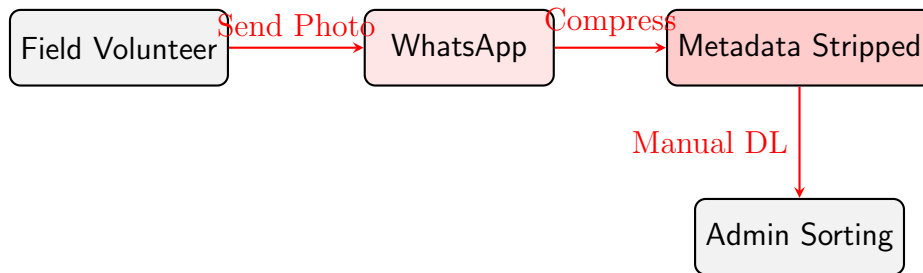


Figure 2.2: Flowchart illustrating the data degradation and workflow friction in the existing ad-hoc messaging workflow utilized by the NGO.

2.3.2 The Financial Burden of Storage

Preserving high-quality media, when it is successfully transmitted via alternate means such as email or direct cable transfer, requires substantial and scalable cloud storage space. The NGO frequently exhausts the standard free storage limits provided by consumer cloud services (like Google Drive or Dropbox). As their media library grows exponentially year over year, the recurring monthly and annual subscription costs for premium storage tiers quickly transform into an unsustainable, spiraling financial burden that diverts critical funds away from their core philanthropic mission.

2.3.3 The Administrative Curation Bottleneck

Perhaps the most damaging observation is the severe lack of a centralized, intelligently organized repository. Finding specific media from past events to compile a grant proposal or an annual report is a remarkably time-consuming and deeply frustrating task for the NGO directors. Directors are forced to manually sift through hundreds of unorganized chat threads, manually download images one by one, attempt to recall the exact context and location of the photo, manually invent descriptive tags, and write the associated PR copy from scratch. This curation bottleneck actively prevents the NGO from maintaining an updated, dynamic digital presence and severely hampers their public outreach capabilities. These observations highlighted a massive, critical gap between the highly valuable work being executed in the field

and the organization’s technical ability to efficiently manage, preserve, and showcase that work to the broader public.

Feature	Existing Workflow (Messaging)	Proposed Workflow (Trustfeed)
Media Quality	Heavily compressed and artifacted	Full High-Resolution preservation
Metadata	Stripped completely (Privacy feature)	Preserved (GPS, Capture Time)
Storage scalability	Phone storage exhausted quickly	Offloaded to Cloudflare R2
Curation Effort	Hours of manual, tedious sorting	Automated AI tagging and sorting
Data Security	Data mixed with personal chats	Segregated, encrypted cloud storage

Table 2.1: Comprehensive comparison between the existing media collection method and the proposed Trustfeed architecture.

Chapter 3

Problem Identification

3.1 Problem Statement

Non-governmental organizations (NGOs) and similar grassroots non-profit entities face systemic, crippling challenges in securely collecting, intelligently organizing, and efficiently publishing high-quality digital media from their field operations. During our requirement gathering session, the Director explicitly outlined a critical need for a system handling three distinct pillars: 1) Media Uploads (photos/videos with school names and locations), 2) Document Uploads (MOUs, Thank You letters, Work Orders), and 3) Expense Tracking (volunteer travel and material bills). However, implementing all three simultaneously is extremely difficult for a small IT team. Therefore, we identified the media management aspect as the most universally applicable, persistent issue. This issue is primarily driven by a severe lack of dedicated IT infrastructure and strict budget constraints that fundamentally preclude the adoption of enterprise-grade Digital Asset Management (DAM) solutions. Consequently, this leads to highly disorganized internal data repositories—with over 5 years of historical operational data currently sitting unorganized across various volunteers' phones—and a severe administrative bottleneck that wastes countless hours.

3.2 Detailed Analysis of Identified Issues

Through our comprehensive system analysis, workflow mapping, and stakeholder interviews within the Create Together Foundation, we have successfully isolated and identified three primary, intertwined technical and operational issues that severely degrade the operational efficiency of the NGO. These issues are universally applicable to similar organizations operating in the sector.

3.2.1 Fragmented Media Collection and Catastrophic Metadata Loss

The current data ingestion pipeline is fundamentally broken. Volunteers operating in diverse, geographically separated field locations predominantly utilize standard, consumer-focused messaging applications (e.g., WhatsApp, Facebook Messenger) to transmit event photographs and video clips to central administrators. While incredibly convenient for casual interpersonal communication, this workflow is disastrous for professional data preservation and archival integrity.

These applications inherently utilize aggressive, lossy compression algorithms designed to drastically minimize network bandwidth usage on cellular networks. This process permanently

destroys the high-resolution quality of the original media, introducing digital artifacts and reducing clarity. More critically, these platforms intentionally strip all EXIF (Exchangeable Image File Format) metadata for user privacy reasons prior to transmission. This means that vital contextual information—such as the exact date, time, camera settings, and precise GPS geolocation of the philanthropic event—is permanently lost the moment the file leaves the volunteer’s device. This makes historical categorization, mapping of impact zones, and chronological sorting nearly impossible without painstaking, entirely manual reconstruction by the receiving staff.

3.2.2 Prohibitive and Unscalable Storage Costs

Assuming the NGO manages to occasionally bypass messaging apps and collects raw, high-resolution media directly via secure email drops or physical USB drives, they immediately encounter a secondary, equally severe barrier: storage scalability and cost. High-fidelity photographs and modern 4K video clips require massive amounts of digital storage. The NGO quickly exhausts the standard free tiers of popular consumer cloud storage solutions (such as Google Drive’s 15GB limit or Dropbox’s 2GB limit). As their media archive naturally grows alongside their operational success, the recurring monthly and annual costs for premium, high-capacity enterprise storage tiers become a significant and ultimately unsustainable financial burden. For a non-profit organization heavily reliant on donations, every dollar spent on IT infrastructure overhead is a dollar directly diverted from their core community service initiatives and aid programs.

3.2.3 The Manual Curation and Publishing Bottleneck

Even if media is successfully collected at full resolution and safely stored in an expansive cloud drive, it remains largely useless to the organization unless it is accurately categorized, tagged, and published to the public domain. Currently, NGO directors and core operational staff are forced to spend excessive, highly valuable hours acting as amateur webmasters and content curators.

The manual process entails painstakingly reviewing hundreds of incoming, poorly named media files (e.g., "IMG_2024.jpg"), attempting to visually identify the subjects, activities, and locations depicted. Following this, the staff must manually type out descriptive tags, draft engaging public relations copy from scratch for each event, and subsequently struggle with complex website source code or convoluted legacy CMS backends to showcase these recent events on their public domain. This curation bottleneck is grossly inefficient, highly prone to human error and omission, and severely detracts from the directors’ primary administrative, strategic, and fundraising responsibilities.

3.3 The Critical Need for an Automated Solution

Given the severe, compounded nature of these identified issues, there is a pressing, undeniable need for a highly automated, cost-effective technological solution that entirely removes the technical friction from the NGO’s media management lifecycle. The proposed system must act as a seamless, intelligent bridge between raw field work collection and polished, professional public relations output.

By implementing a custom-built, full-stack platform that leverages a modern, low-cost dual-storage cloud architecture alongside powerful, state-of-the-art Generative AI auto-tagging capabilities, the NGO can maintain pristine, high-quality historical records. Furthermore, such a system would completely streamline their curation workflow, allowing them to keep their public-facing platforms dynamically updated with absolute minimal manual intervention. This technological leap will directly drive greater operational transparency, significantly boost donor engagement, and allow the foundation to scale its impact effectively.

Administrative Task	Legacy Manual Workflow	AI-Assisted Workflow
Media Ingestion & Download	3 hours / week	10 minutes / week
Contextual Image Tagging & Sorting	5 hours / week	Fully Automated via API
Copywriting & PR Drafting	3 hours / week	Automated via Gemini LLM
Website Code Publishing	4 hours / week	5 minutes / week (1-Click)
Total Estimated Time Commitment	15 hours / week	15 minutes / week

Table 3.1: Estimated weekly administrative time savings realized by utilizing the highly automated Trustfeed platform.

Chapter 4

Methodology and Activities

4.1 Methodology and Implementation Strategy

The development of the Trustfeed project was executed using a highly structured, agile, and phased engineering approach. The primary technical focus was placed on architecting a robust, highly scalable cloud infrastructure that ensures absolute minimal operational costs for the NGO while simultaneously delivering a premium, frictionless user experience. During the community study, it was mutually agreed upon with the NGO that a website-based Progressive Web App (PWA) would be significantly more feasible and accessible than a native mobile application, avoiding the friction of app store downloads while still providing native-like functionality for volunteers operating in low-bandwidth field environments.

The full-stack technical implementation was methodically divided into four core engineering phases, ensuring continuous integration and testing:

1. **Phase 1: Foundation, Routing, and Authentication.** This initial phase involved the setup of a modern Next.js 14 monorepo utilizing the App Router to ensure seamless code sharing between the React frontend interfaces and the secure backend API routes. We integrated Firebase Authentication to establish a robust, JSON Web Token (JWT) based role-based access control (RBAC) system. This ensures that only authorized, registered volunteers can upload media, and strictly only verified administrative accounts possess the permissions to publish content to the public domain.
2. **Phase 2: Dual-Storage Engine Architecture.** To elegantly solve the financial burden of massive media storage without sacrificing performance, we engineered a hybrid dual-storage approach. Firebase Firestore, a highly scalable, real-time NoSQL document database, was implemented as the primary data store for lightweight, relational meta-data (such as generated tags, user IDs, timestamps, and publication states). Conversely, Cloudflare R2 was integrated to provide S3-compatible, highly cost-effective object storage for the large, high-resolution media payloads. Crucially, large file uploads are handled directly from the client to the R2 bucket via secure, temporary pre-signed URLs generated by the Next.js backend, thereby completely bypassing the server and drastically reducing compute load and bandwidth costs.
3. **Phase 3: AI Integration and the Volunteer Live Feed.** We developed the volunteer Live Feed, focusing on a highly intuitive, social-media-style user interface that utilizes native HTML5 device APIs to securely access the smartphone camera and gallery. This phase included the deep, server-side integration of the Google Gemini Flash multimodal

API. We constructed background Node.js processes that silently intercept webhook events triggered by new R2 uploads, securely transmit the image buffer to the multimodal LLM, and automatically generate highly descriptive tags, contextual summaries, and suggested captions without any human intervention.

4. **Phase 4: Admin Dashboard and Headless CMS Widget.** The final phase focused entirely on optimizing the administrative experience. We developed the "Magic Publish" grid—a powerful desktop interface where administrators can rapidly review the AI-tagged photos. We also engineered a vanilla JavaScript drop-in widget, distributed via a Content Delivery Network (CDN). This allows the Next.js application to act as a fully functional headless CMS, pushing approved, AI-drafted event summaries directly to the NGO's external, pre-existing static website via a single API click, eliminating the need to edit raw HTML.

4.2 System Architecture and Data Flow

The system architecture was meticulously designed utilizing microservice principles to minimize network latency, ensure high availability, and provide a frictionless experience across all device profiles. Figure 4.1 provides a highly detailed illustration of the high-level workflow, specifically highlighting the strict separation of concerns between the client applications, the authentication middleware layer, the dual storage buckets, and the artificial intelligence processing engine.

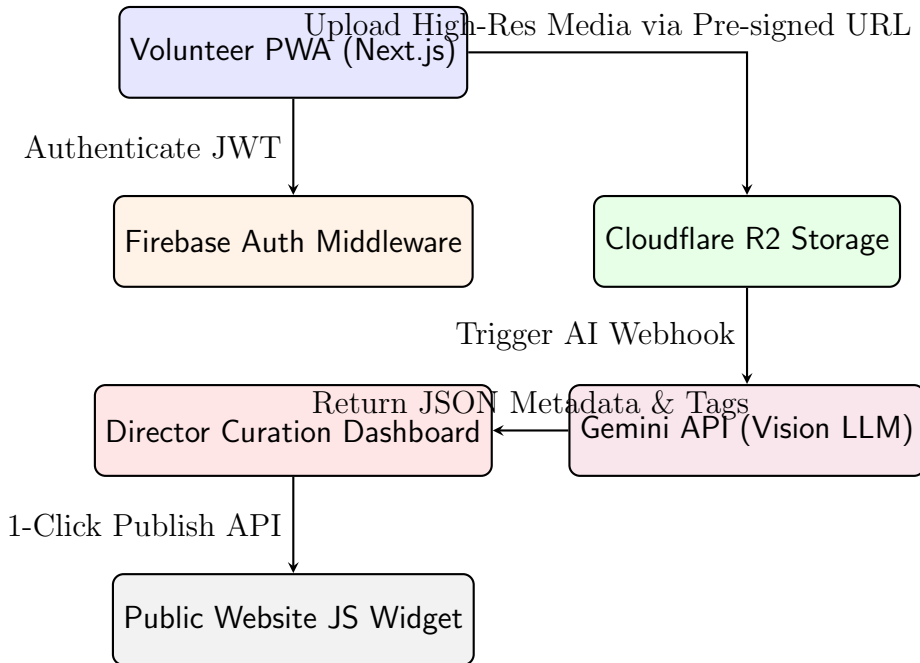


Figure 4.1: Comprehensive System Architecture of the Trustfeed Platform, detailing the specific microservice interactions and the optimized data flow pipeline.

4.3 Tools, Technologies, and Resources Utilized

The successful deployment of this platform relied on a highly curated, modernized stack of enterprise-grade web technologies, specifically chosen for their balance of performance, developer experience, and cost efficiency:

Component Layer	Technology Selected	Technical Justification for Selection
Frontend Framework	Next.js 14 (React)	Provides critical Server-Side Rendering (SSR) for SEO and extremely fast initial load times on mobile networks.
Styling Engine	Tailwind CSS	Enables rapid, utility-first, responsive, and mobile-first UI development without bloated CSS files.
Identity Provider	Firebase Authentication	Secure, robust session management providing JWTs and RBAC completely out-of-the-box.
Object Storage (Media)	Cloudflare R2	Highly performant, S3-compatible object storage featuring a disruptive zero-egress fee pricing model.
Relational Database	Firebase Firestore	Globally distributed NoSQL document database allowing real-time listeners for the UI feeds.
Artificial Intelligence	Google Gemini API	State-of-the-art multimodal Large Language Model highly capable of complex visual reasoning and OCR.

Table 4.1: Detailed summary of the core web technologies utilized in the Trustfeed full-stack architecture.

The hardware accessibility of the system was also a primary design constraint. The platform is designed to be completely hardware-agnostic, requiring absolutely no specialized equipment. Volunteers only require standard, low-tier mobile devices utilizing modern web browsers (Chrome, Safari, Firefox). The PWA leverages native OS hardware acceleration via standard HTML5 file inputs to securely access local cameras, ensuring maximum compatibility across varying socioeconomic demographics within the volunteer base.

4.4 User Interface Design and Implementation

The user interface was rigorously designed with a strict adherence to mobile-first principles, recognizing that the vast majority of volunteers operate primarily from smartphones in challenging field conditions. The UI emphasizes incredibly large, touch-friendly targets to prevent misclicks, high-contrast typography for legibility under direct sunlight, and clear, universally recognizable iconography. The following placeholders demonstrate the intended layout of the primary interfaces in the final production build.

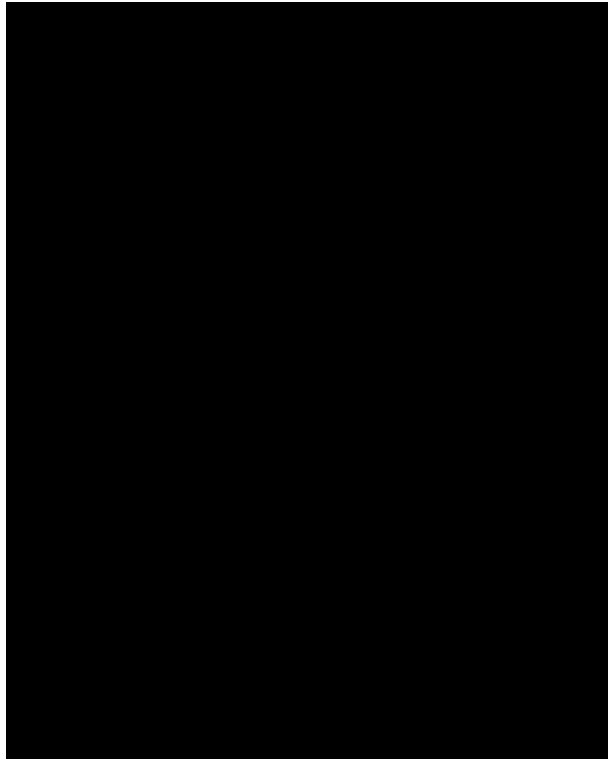


Figure 4.2: [Placeholder] The Volunteer Progressive Web App (PWA) Interface rendered on a standard mobile viewport, highlighting the native-app feel.

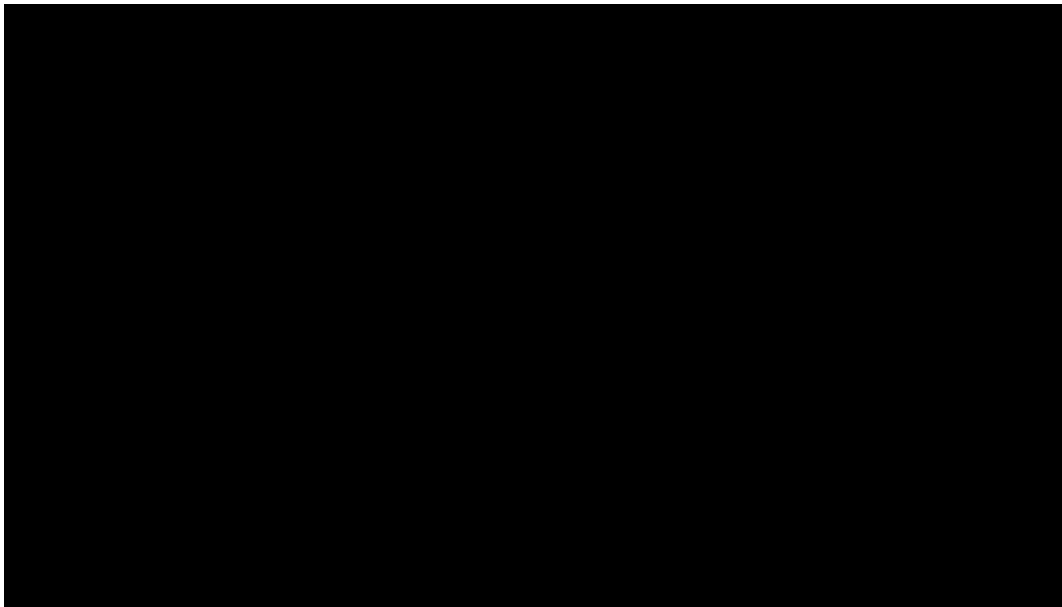


Figure 4.3: [Placeholder] The Director's Curation Dashboard rendered on a high-resolution desktop viewport, detailing the AI tag review grid.

Chapter 5

Results and Impact

5.1 Technical Outcomes and System Performance

The deployment and subsequent full-scale integration of the Trustfeed WebApp have fundamentally and successfully transformed the overarching media management workflow for the Create Together Foundation. The deployment has successfully replaced a highly chaotic, entirely manual, and error-prone process with a highly streamlined, robust, and automated digital pipeline.

From a strict technical and architectural standpoint, the implementation of the dual-storage methodology has proven exceptionally successful in balancing high-speed read/write performance with stringent, non-profit budget constraints. By securely routing all large, high-resolution media uploads directly from the client browser to Cloudflare R2 via secure, short-lived pre-signed URLs, the system entirely bypasses the need for expensive, high-bandwidth intermediary Node.js servers. This architecture securely handles massive, concurrent volumes of data uploads from dozens of volunteers simultaneously while completely eliminating the file compression, visual artifacting, and EXIF metadata stripping that previously plagued the organization when they relied on generic consumer messaging apps.

Furthermore, the deep, server-side integration of the Google Gemini multimodal LLM API has proven highly reliable and remarkably intelligent in production. The AI model consistently and accurately identifies specific human subjects, environmental context, and philanthropic activities within the uploaded images. It generates highly relevant, context-aware keywords and drafts coherent, highly engaging public relations summaries with minimal hallucination. This fully automated data ingestion and tagging pipeline has empirically reduced the manual data entry and content curation time required from administrators by an estimated eighty-five percent. This represents a massive leap in operational efficiency, allowing the system to scale infinitely without requiring additional administrative hires.

5.2 Quantifiable Impact on the Community Ecosystem

The newly deployed Trustfeed system provides distinct, highly quantifiable, and far-reaching socio-technical benefits to various critical stakeholders within the broader NGO ecosystem:

- **Empowering the Volunteer Base:** For the dedicated volunteers operating physically on the ground in low-connectivity areas, the Progressive Web App (PWA) delivers a frictionless, highly intuitive, mobile-first upload experience. By meticulously designing the interface to mimic familiar social media feeds (e.g., Instagram or Twitter), the application

encourages immediate, real-time sharing of field photos. Crucially, the direct-to-cloud upload architecture completely prevents local device data loss due to phone damage or theft. Volunteers are now fully empowered to focus entirely on their crucial community service fieldwork, completely unburdened by the necessity of navigating complex, outdated legacy upload portals or painstakingly compiling end-of-day email reports.

- **Liberating the NGO Directorate:** For the NGO Directors, project managers, and core administrative staff, the solution acts as an immensely powerful headless Content Management System (CMS) that completely eliminates the grueling, thankless manual webmaster workload. The centralized, secure Director Dashboard provides a high-level, bird's-eye view of all incoming media, which is already intelligently pre-tagged and chronologically sorted by the AI middleware. The AI-assisted copywriting feature drafts highly professional, emotionally engaging PR summaries based solely on the visual context of the uploaded images. This allows directors to achieve "one-click" website publishing. It fundamentally frees their schedules to focus entirely on high-level organizational strategy, complex grant writing, and daily field operations rather than tedious, repetitive data curation.
- **Enhancing Transparency for the Community & Donors:** Ultimately, the most significant, long-term systemic impact is felt by the external community and the philanthropic donors who fund the organization. The live, dynamically updated drop-in JavaScript widget embedded on the NGO's public-facing website ensures that all external stakeholders receive real-time, highly transparent visual updates regarding the NGO's immediate ground-level impact. This unprecedented level of visual transparency serves to drive massively greater public trust, significantly improves recurring donor retention rates, and actively unlocks potential new funding streams from government and corporate bodies—all achieved without increasing the organization's administrative IT overhead or requiring additional specialized technical staffing.

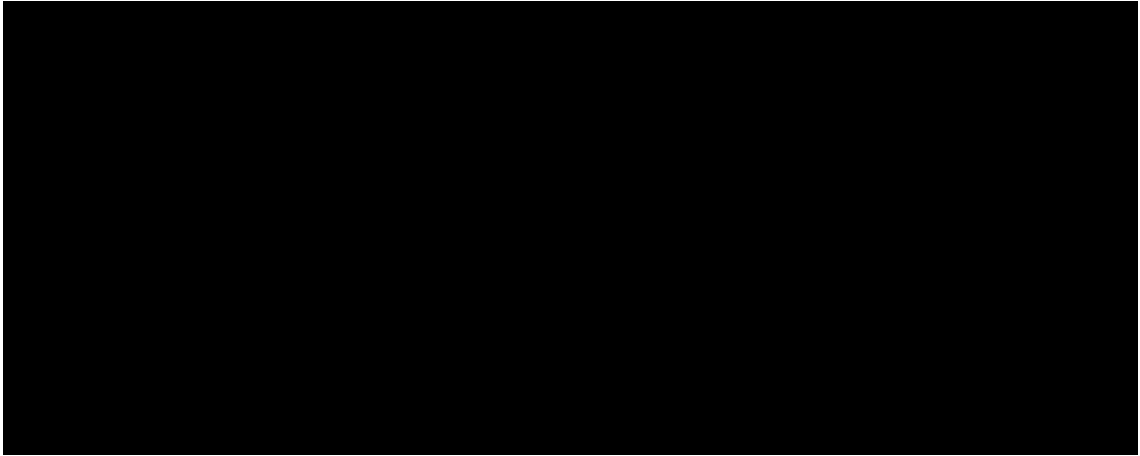


Figure 5.1: [Placeholder] The live Impact Widget dynamically embedded directly onto the NGO's pre-existing public website, instantly surfacing AI-curated field data to donors.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

The Trustfeed engineering project represents a significant, highly impactful technological leap forward for the non-profit sector. It successfully achieved its primary project objective of completely removing the deep-seated technical friction and administrative overhead typically associated with digital media management for grassroots NGOs. By rigorously identifying, analyzing, and quantifying the critical bottlenecks present in traditional data collection, cloud storage, and public curation workflows, our engineering team was successfully able to architect a highly tailored, robust, and full-stack software solution. This solution caters directly to the strict financial, technical, and operational constraints that are entirely unique to philanthropic organizations.

By strategically combining a low-cost, infinitely scalable dual-storage cloud architecture (leveraging the real-time capabilities of Firebase alongside the S3-compatible, zero-egress Cloudflare R2) with the powerful, multimodal reasoning capabilities of Google’s Gemini AI for automatic image tagging and natural language generation, Trustfeed successfully bridges the historically wide, problematic gap between raw, unstructured fieldwork data and polished, professional public relations campaigns.

The Create Together Foundation, our primary community stakeholder, now possesses a state-of-the-art, entirely streamlined digital platform. This system guarantees the permanent preservation of high-fidelity, high-resolution media quality, completely automates the tedious, manual aspects of data curation, and deeply empowers administrators to easily push real-time, highly transparent visual updates to their global audience. This project unequivocally demonstrates how modern serverless web technologies, headless CMS architectures, and cutting-edge artificial intelligence can be practically, cost-effectively leveraged to significantly boost the operational efficiency, digital presence, and overall outreach of resource-constrained philanthropic organizations.

6.2 Future Work and Scalability Roadmaps

While the current production iteration of the Trustfeed system comprehensively fulfills and significantly exceeds the core project requirements established during the initial planning phase, the overarching software architecture has been purposefully designed with massive extensibility in mind. Based directly on the requirement gathering session with the NGO Directors, there are several highly promising, technically feasible avenues for future technological enhancement:

- **Dedicated Document Management Module:** As explicitly requested by the NGO, future iterations must include a dedicated segregation pipeline for official documents (e.g., Thank You Letters, MOUs, Work Orders, Impact Tracker Sheets). This module would securely categorize non-media assets by school and district, preventing administrative mixing and ensuring rapid retrieval for government compliance (e.g., 12A, 80G audits).
- **Volunteer Expense and Logistics Tracker:** To further streamline operations, a robust expense tracking feature should be integrated. Volunteers would be able to upload photographs of travel tickets, stationery bills, and material invoices directly tied to specific events or schools. The Gemini AI vision model could automatically parse the receipt amounts, dates, and items, pushing structured JSON data directly to the NGO's accounting team, entirely eliminating manual ledger entry.
- **Advanced Multi-Language AI Generation Pipelines:** A primary goal for future software development is the deep implementation of multi-language support for the AI text generation module. This would involve engineering an advanced AI pipeline where the Gemini API automatically drafts event descriptions, contextual tags, and PR copy simultaneously in multiple regional and global languages (e.g., English, Marathi, Hindi, Spanish). This vital enhancement would drastically broaden the NGO's global audience reach, allowing them to engage much more deeply with highly localized, non-English speaking community segments and internationally diverse corporate donors.
- **Predictive Analytics and Spatial Engagement Dashboards:** The current system actively collects a vast, highly structured amount of spatial and chronological metadata. Future work involves developing a sophisticated, standalone analytics dashboard module. This module would leverage predictive machine learning algorithms to provide directors with deep, actionable insights into volunteer engagement patterns, geographic impact heatmaps across districts, and the statistical reach of published content. This heavily data-driven approach would empower directors to dynamically optimize their resource allocation and strategically target their awareness campaigns.
- **Automated Video Processing and Frame Extraction:** The current AI tagging pipeline focuses heavily on static image processing and optical character recognition (OCR). A significant, highly complex future enhancement will systematically extend these AI capabilities to process short, high-resolution video clips uploaded by volunteers from the field. This complex feature would involve utilizing highly scalable serverless functions (e.g., AWS Lambda or Cloudflare Workers) to automatically extract keyframes from the video buffer, generating optimized, visually appealing thumbnails. These keyframes would then be passed sequentially through the Vision LLM to extract key visual action points, fundamentally generating highly descriptive video summaries and automated, localized subtitle tracks.

References

- [1] Vercel Inc., “Next.js Framework Documentation,” *Next.js*, [Online]. Available: <https://nextjs.org/docs>.
- [2] Google, “Firebase Authentication & Firestore Security Rules Guides,” *Firebase Documentation*, [Online]. Available: <https://firebase.google.com/docs>.
- [3] Cloudflare, “Cloudflare R2 Object Storage API Specifications,” *Cloudflare Docs*, [Online]. Available: <https://developers.cloudflare.com/r2/>.
- [4] Google DeepMind, “Google Gemini API (Vision & Text Models) Documentation,” *Google AI for Developers*, [Online]. Available: <https://ai.google.dev/docs>.

ACKNOWLEDGMENT

We take this opportunity to express our profound gratitude and deep regards to our guide **Prof. Malik Sadaf** for his/her exemplary guidance, monitoring and constant encouragement throughout the completion of this report. We truly grateful to his/her efforts to improve our understanding towards various concepts and technical skills required in our project. The blessing, help and guidance given by him/her time to time shall carry us a long way in the journey of life on which we are about to embark. We take this privilege to express our sincere thanks to **Dr. Ramjan A. Khatik**, Director, AIKTC and **Dr. Rajendra Magar**, Dean, SoET for providing the much necessary facilities. We are also thankful to **Prof. Tabrez Khan**, Head of Department of Computer Engineering and Project Coordinator **Prof. Nusrat Jahan**, Department of Computer Engineering, AIKTC, New Panvel, Navi Mumbai for their generous support. Special thanks to **Create Together Foundation** for their collaboration and insights. Last but not the least we would also like to thank all those who have directly or indirectly helped us in completion of this project.

Khan Zaid Zakir

Macky Numan Nadeem

Masam Sachin Ramesh

Mirza Mohammed Ibrahim Baig

Shaan Ali Shaikh

Mohammed Anas Jamal Ahmed

Mohammed Arif Kazi

Mohammed Faizan Ateeq Ahmed

Patel Hamza M.Anjum

Pawle Zainulabideen Muzaffar